

2015 GCSE Syllabus Worksheets

Foundation Worksheets

Grades 1 to 5

Grade 1

Grade 2

Grade 3

Grade 4

Grade 5

Part 3

Instructions for use

Clicking on something like this **Grade 4** will take you to the Grade 4 clip numbers and titles.



Clicking on something like this Compound Units. 142 will take you to Worksheet 142.



Clicking on something like this

©MathsWatch	Clip 142
Grade 4 questions	

 will take you back to the contents page.



Take me to . . .

Grade 1

Grade 2

Grade 3

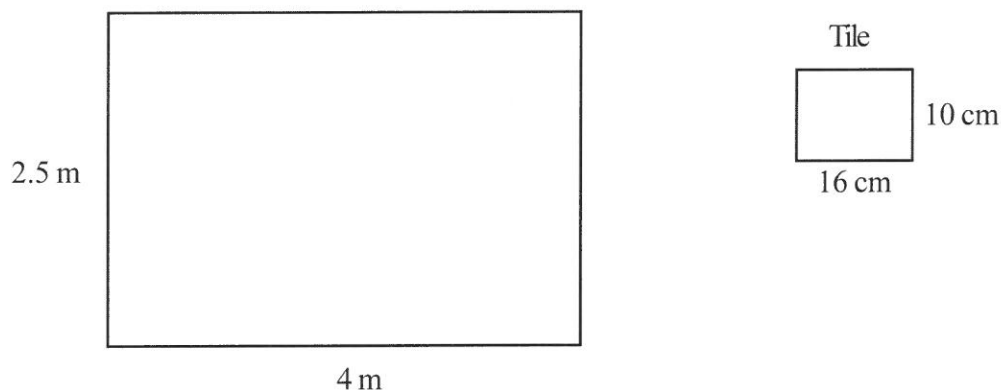
Grade 4

Grade 5

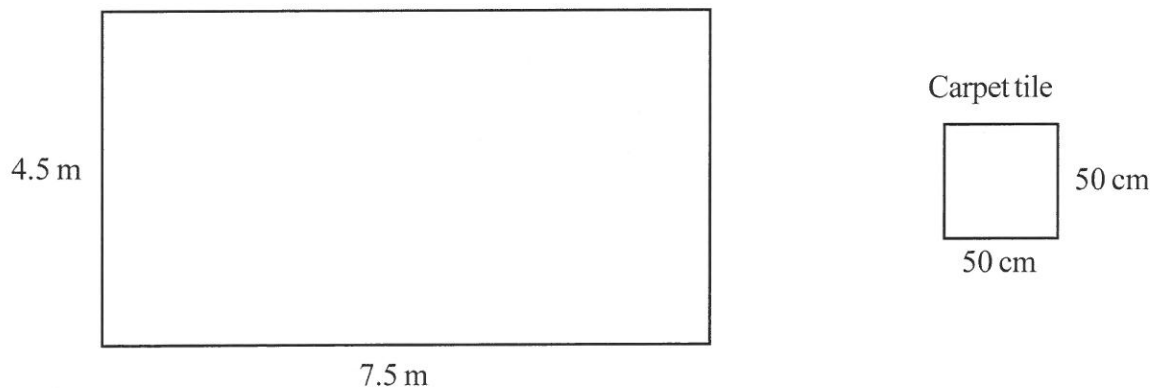
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- 1) Change 9 m^2 into cm^2
- 2) How many square metres are there in 5 square kilometres?
- 3) Change 4 cm^2 into mm^2
- 4) Convert 6.5 m^2 into mm^2
- 5) Change 2 m^3 into cm^3
- 6) How many cubic millimetres are there in 3 cubic centimetres?
- 7) Change 7 m^3 into mm^3
- 8) A tiler wants to tile a rectangular wall which measures 4 m by 2.5 m.
Each tile measures 16 cm by 10 cm.
How many tiles will he need for the wall?



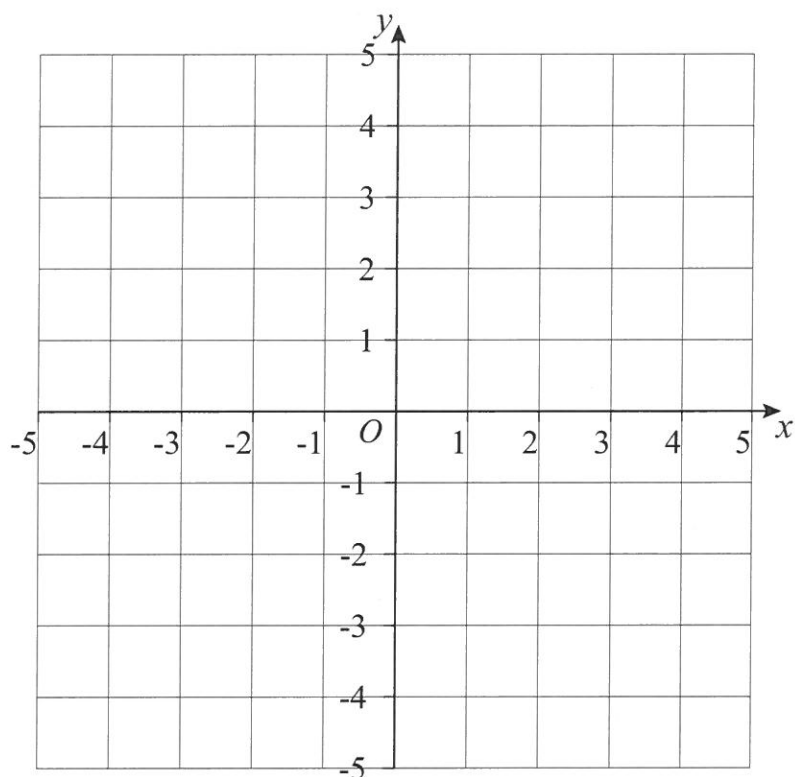
- 9) A carpet-fitter is laying carpet tiles on a rectangular floor which measures 7.5 m by 4.5 m.
Each carpet tile measures 50 cm by 50 cm.
How many carpet tiles will he need for the floor?



Problems on Coordinate Axes

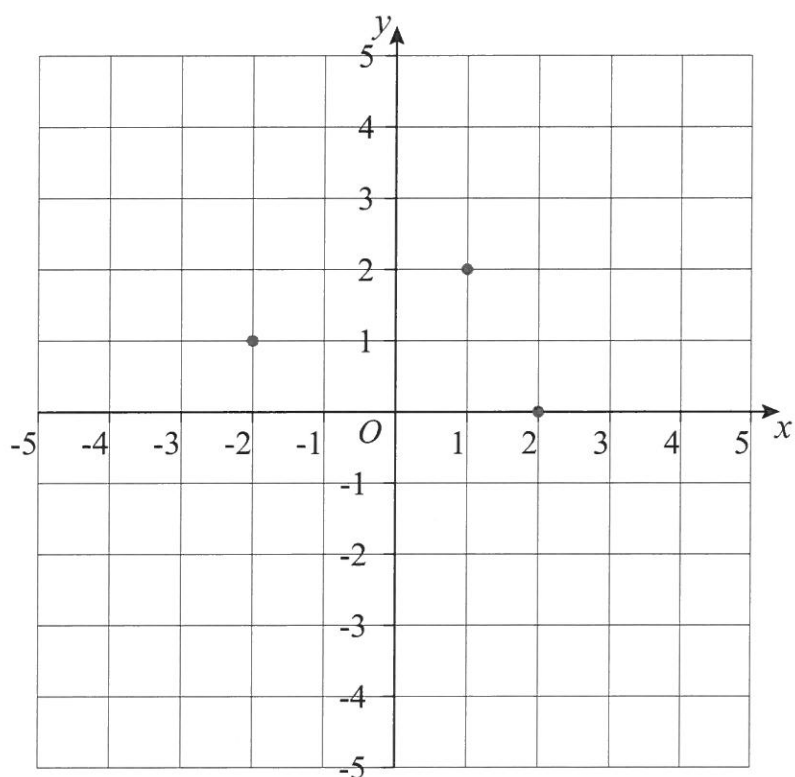
- 1) The points $A(3, 0)$, $B(2, 4)$ and $C(-3, 1)$ are three corners of a parallelogram.

What are the coordinates of the 4th corner?

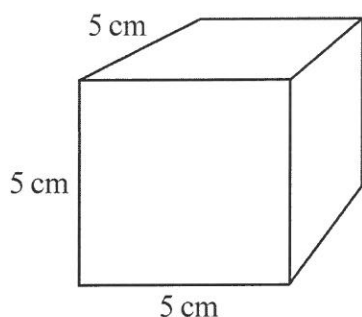


- 2) The diagram shows 3 vertices of a parallelogram.

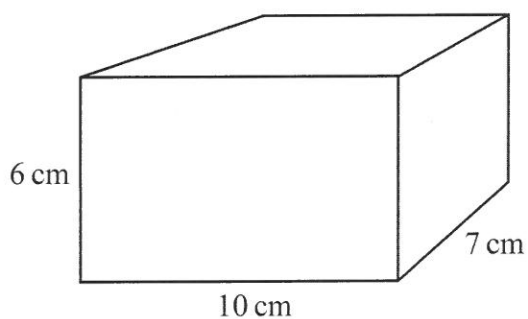
Write down the coordinates of all the possibilities for the 4th vertex.



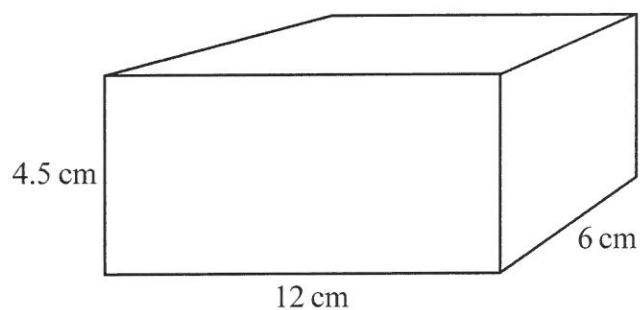
- 1) A cube has sides of length 5 cm.
Find the total surface area of the cube.



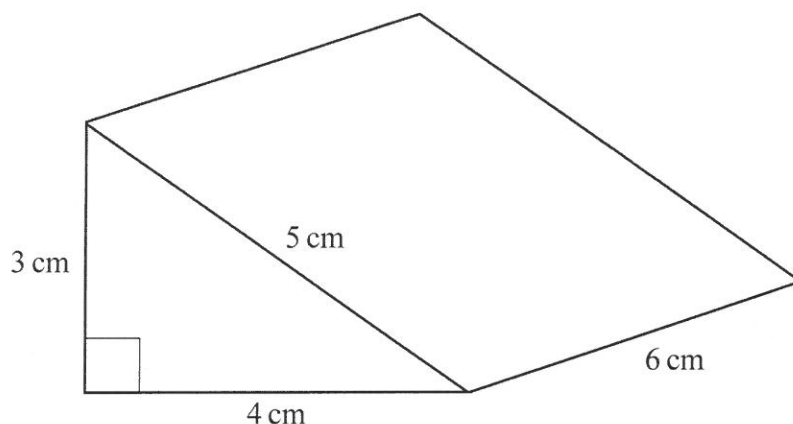
- 2) A cuboid has sides of length 10 cm, 6 cm and 7 cm.
Find the total surface area of the cuboid.



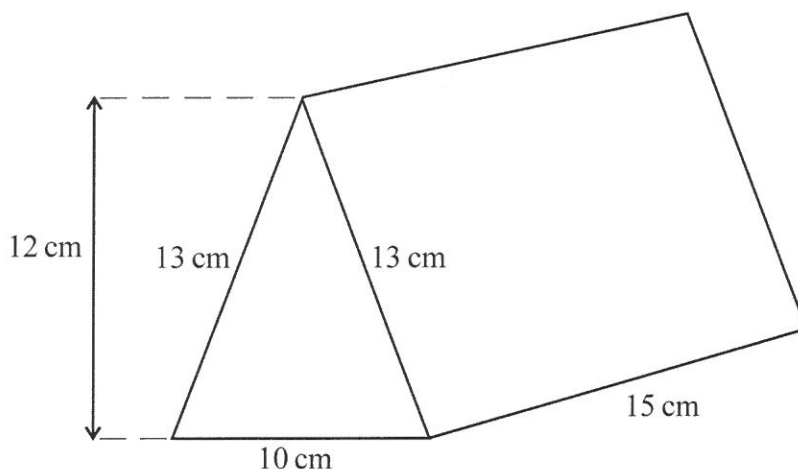
- 3) A cuboid has sides of length 12 cm, 4.5 cm and 6 cm.
Find the total surface area of the cuboid.



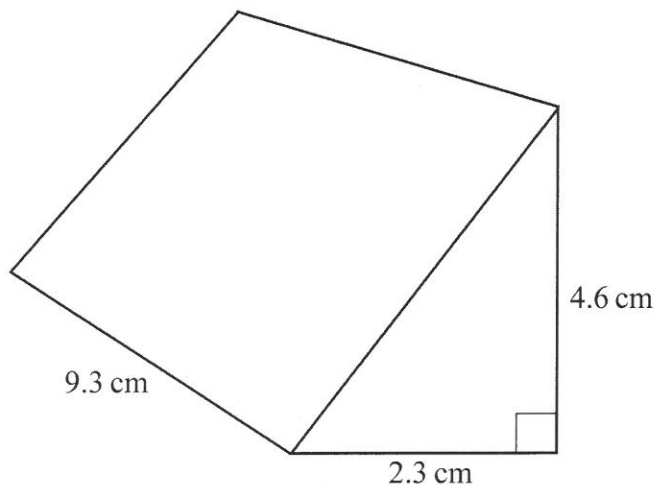
- 1) Find the surface area of this triangular prism.



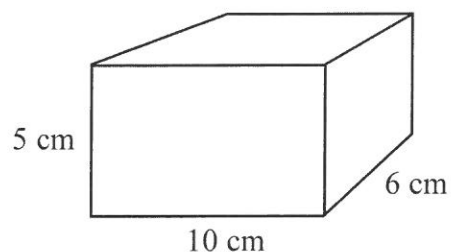
- 2) Find the surface area of this triangular prism.



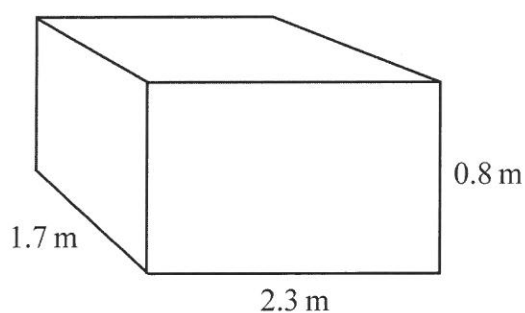
- 3) With the aid of Pythagoras' Theorem, find the surface area of this triangular prism.
Give your answer correct to 2 significant figures.



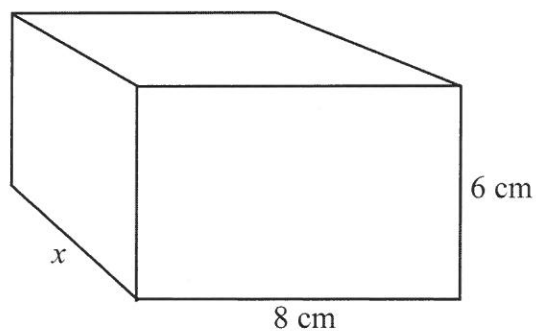
- 1) Find the volume of this cuboid.



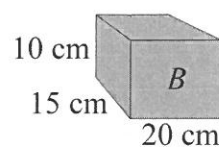
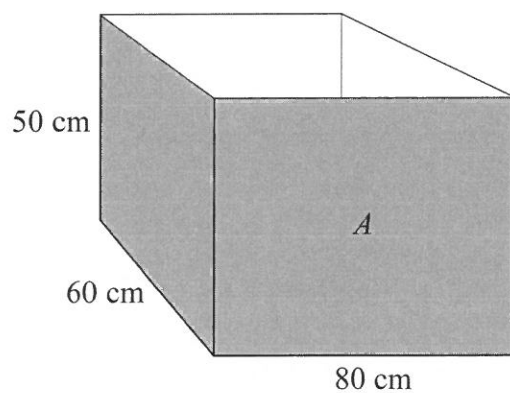
- 2) Find the volume of this cuboid.



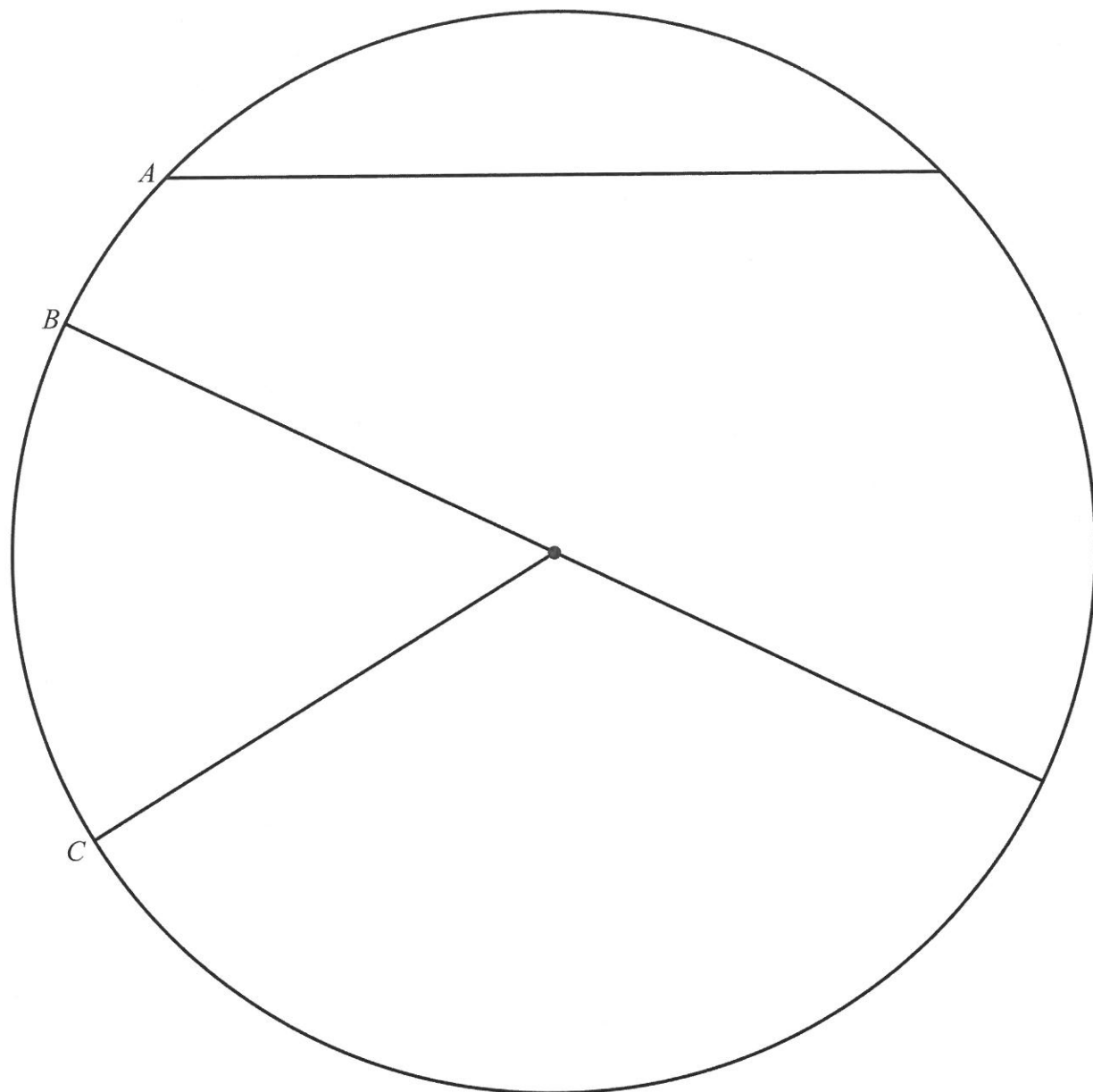
- 3) The volume of this cuboid is 480 cm^3 .
Find the length of the side marked x .



- 4) Boxes A and B are both cuboids.
How many of box B could be packed into box A ?



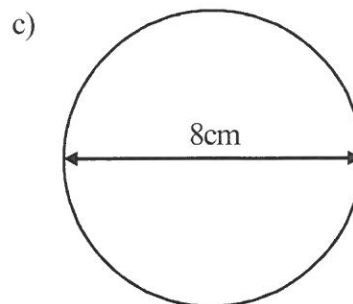
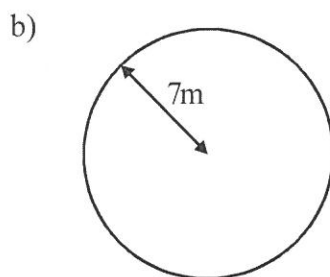
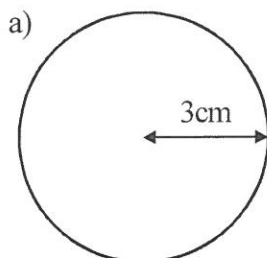
- 1) In the circle, write the correct names for line A , B and C .



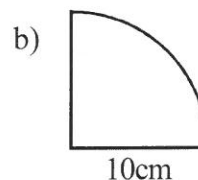
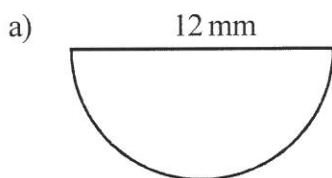
- 2) What is the special name given to the perimeter of a circle?



1) Find the areas of the following shapes.



2) Work out the areas of the following shapes.



3) The **radius** of the top of a circular table is 60 cm.
The table also has a circular base with **diameter** 30 cm.

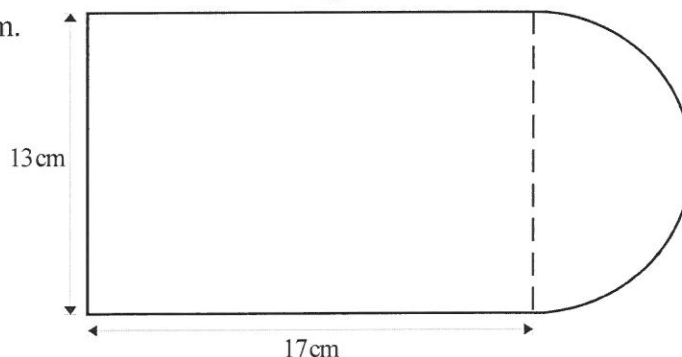
a) Work out the area of the top of the table.

b) Work out the area of the base of the table.



4) The diagram shows a shape, made from a semi-circle and a rectangle.
The diameter of the semi-circle is 13 cm.
The length of the rectangle is 17 cm.

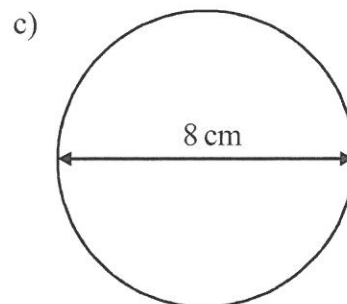
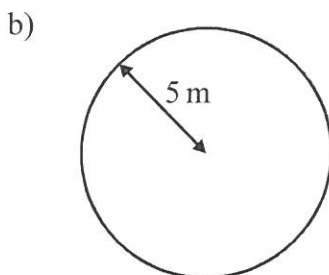
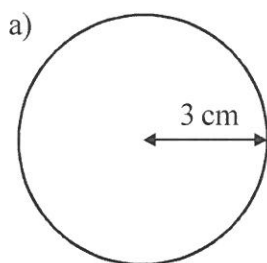
Calculate the area of the shape.
Give your answer correct to
3 significant figures.



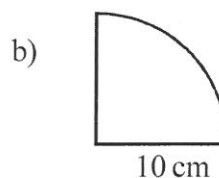
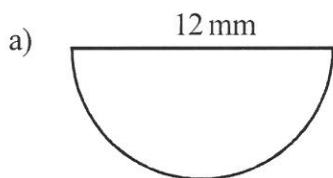
Circumference of a Circle



- 1) Find the circumference of the following shapes.



- 2) Work out the perimeter of the following shapes.



- 3) The **radius** of the top of a circular table is 60 cm.
The table also has a circular base with **diameter** 30 cm.

a) Work out the circumference of the top of the table.

Let π be 3.14

b) Work out the circumference of the base of the table.

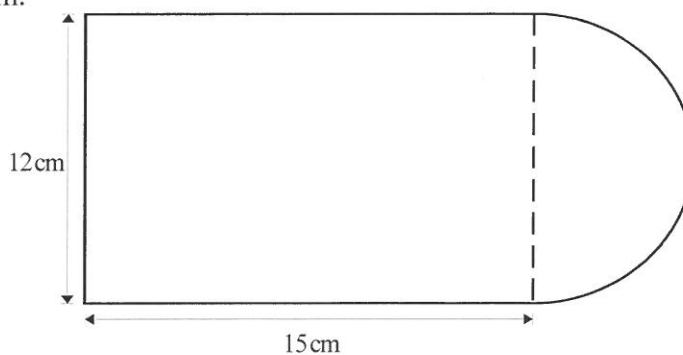
Let π be 3.14



- 4) The diagram shows a shape, made from a semi-circle and a rectangle.
The diameter of the semi-circle is 12 cm.
The length of the rectangle is 15 cm.

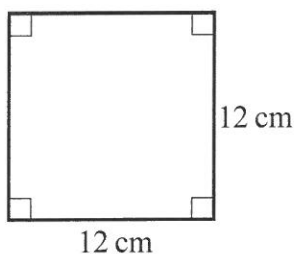
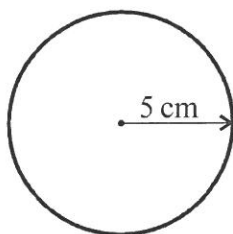
Calculate the perimeter of the shape.

Give your answer correct to
3 significant figures.



Area and Circumference of a Circle

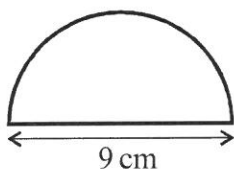
- 1) A circle has a radius of 5 cm.
A square has sides of length 12 cm.



Work out the difference between the area of the circle and the area of the square if you take π to be 3.



- 2) Here is a tile in the shape of a semi-circle.



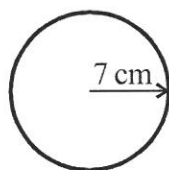
The diameter of the semi-circle is 9 cm.

Work out the perimeter of the tile.

Give your answer correct to two decimal places.



- 3) A circle has a radius of 7 cm.

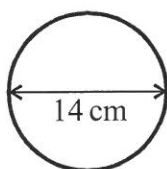


Work out the area of the circle.

Give your answer correct to three significant figures.



- 4) A circle has a diameter of 14 cm.

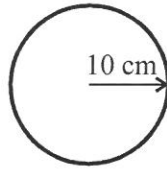


Work out the circumference of the circle.

Give your answer correct to three significant figures.



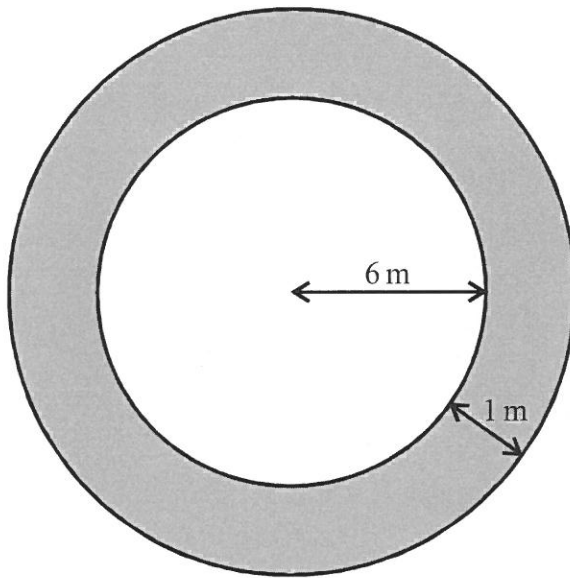
- 1) The radius of a circle is 10 cm.



Work out the area of this circle.



- 2) The diagram shows a circular pond with a path around it.



The pond has a radius of 6 m.

The path has a width of 1 m.

Work out the area of the path.

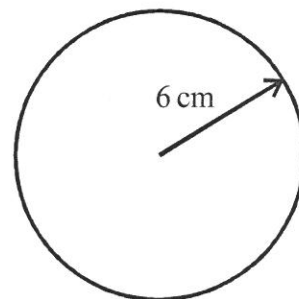
Give your answer correct to 3 significant figures.



- 3) The diagram shows a CD which has a radius of 6 cm.

- a) Work out the circumference of the CD.

Give your answer correct to 3 significant figures.



CDs of this size are cut from rectangular sheets of plastic.

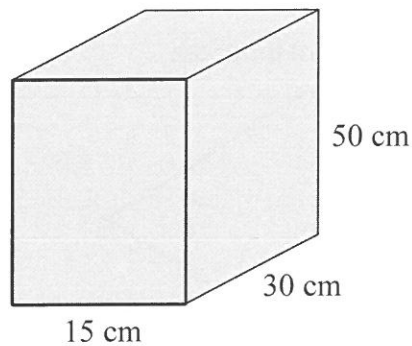
Each sheet is 1 metre long and 50 cm wide.

- b) Work out the greatest number of CDs which can be cut from one rectangular sheet.

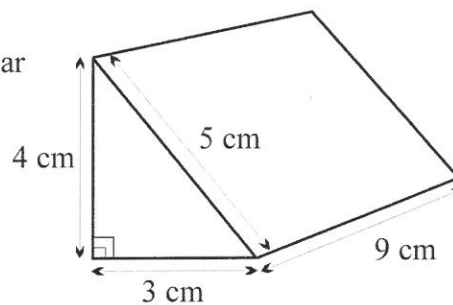
Volume of a Prism



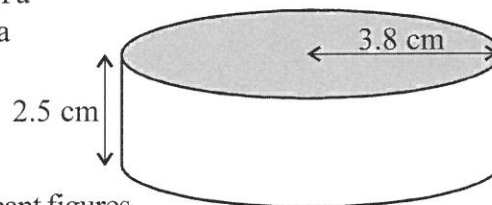
- 1) The diagram shows a cuboid.
Work out the volume of the cuboid.



- 2) Calculate the volume of this triangular prism.



- 3) An ice hockey puck is in the shape of a cylinder with a radius of 3.8 cm and a thickness of 2.5 cm.



Work out the volume of the puck.
Give your answer correct to 3 significant figures.

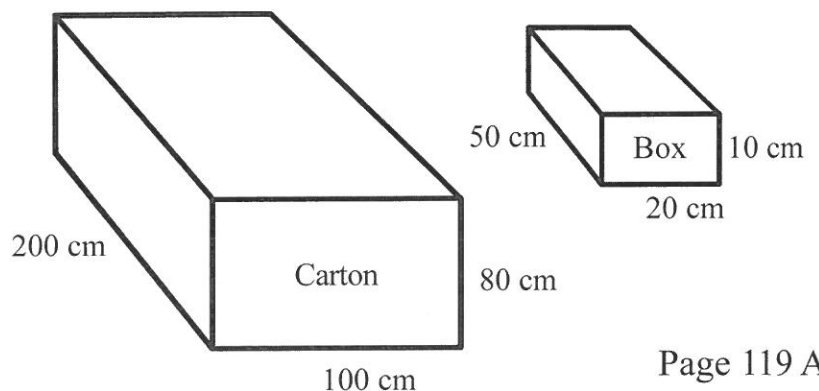


- 4) A cuboid has: a volume of 80cm^3
a length of 5 cm
a width of 2 cm

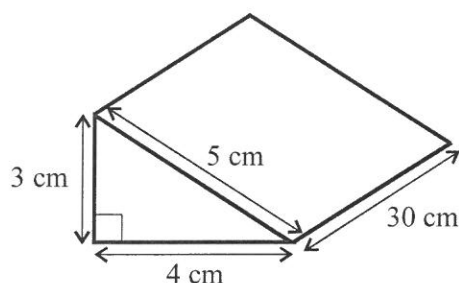
Work out the height of the cuboid.



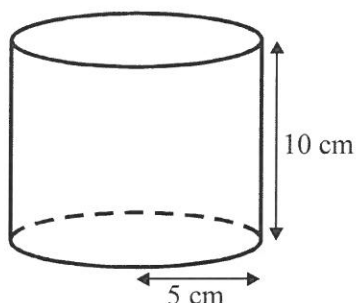
- 5) Work out the maximum number of boxes which can fit in the carton.



- 1) Work out the volume of the prism.



- 2)

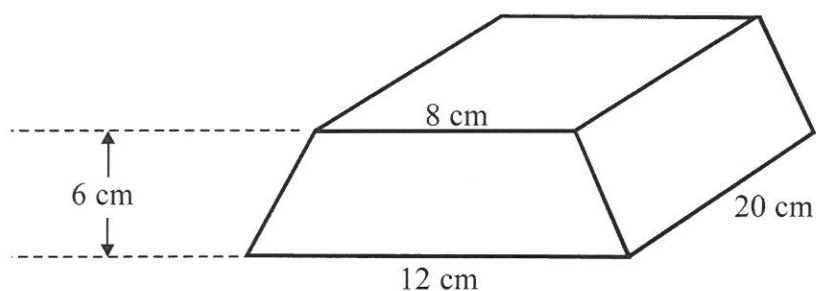


A solid cylinder has a radius of 5 cm and a height of 10 cm.

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

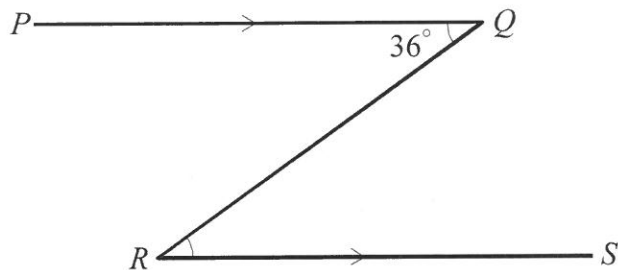
- 3)



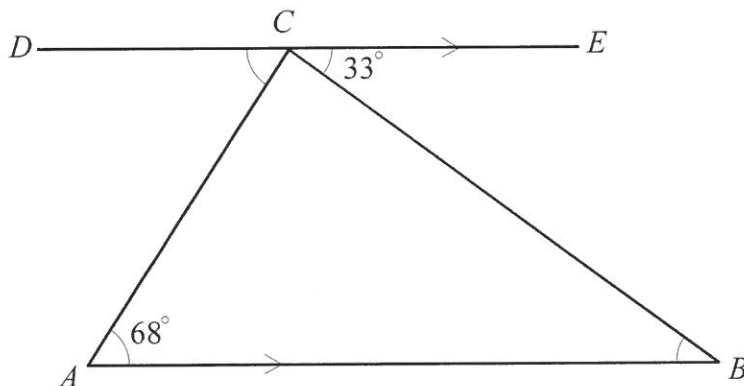
The diagram shows a solid prism made from metal.
The cross-section of the prism is a trapezium.

Find the volume of the prism.
You must state your units.

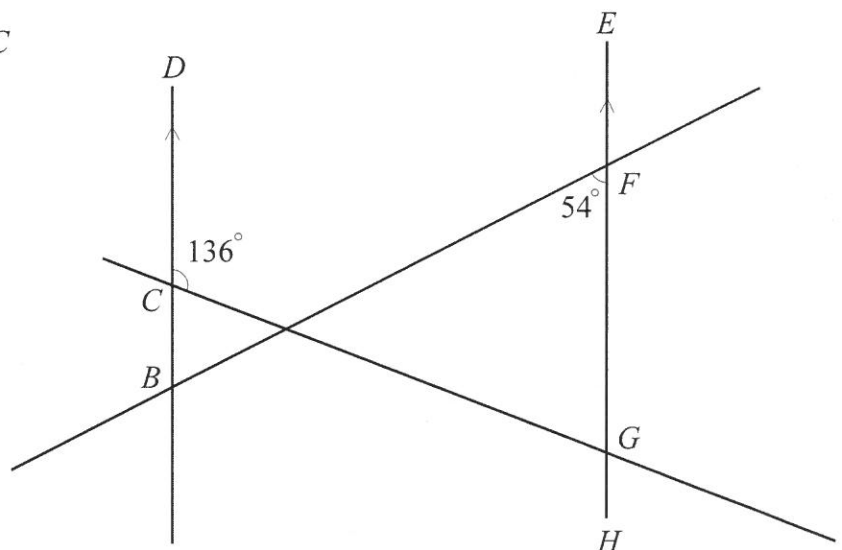
- 1) Line PQ is parallel to line RS .
 If angle PQR is equal to 36°
 a) What is the size of angle QRS ?
 b) Give a reason for your answer.



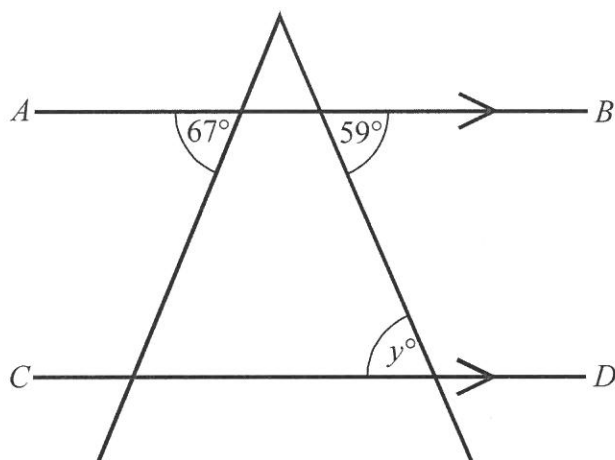
- 2) Line DCE is parallel to line AB
 a) Find the size of angle ABC
 b) Find the size of angle DCA
 c) Calculate the size of angle ACB



- 3) a) Find the size of angle DBF
 b) Find the size of angle HGC



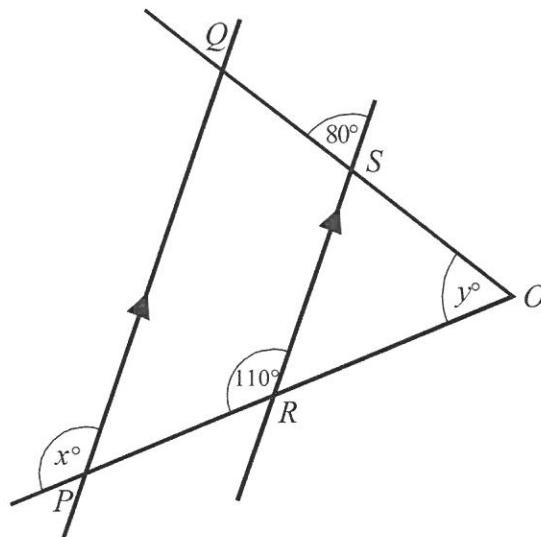
1)



AB is parallel to CD .

- Write down the value of y .
- Give a reason for your answer.

2)



PQ is parallel to RS .

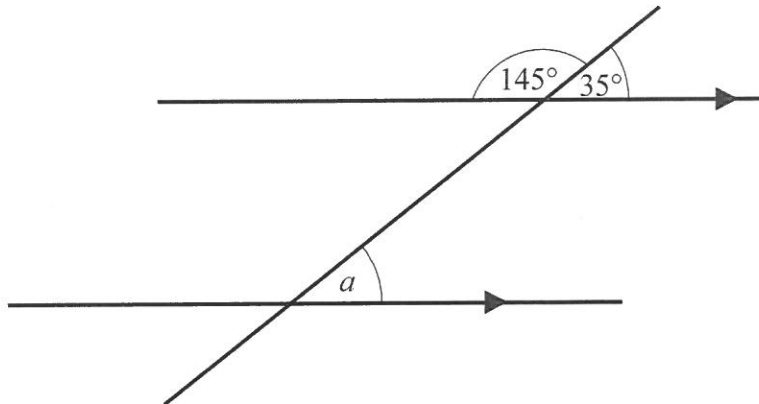
OSQ and ORP are straight lines.

- Write down the value of x .
 - Give a reason for your answer.

- Work out the value of y .

Angles and Parallel Lines

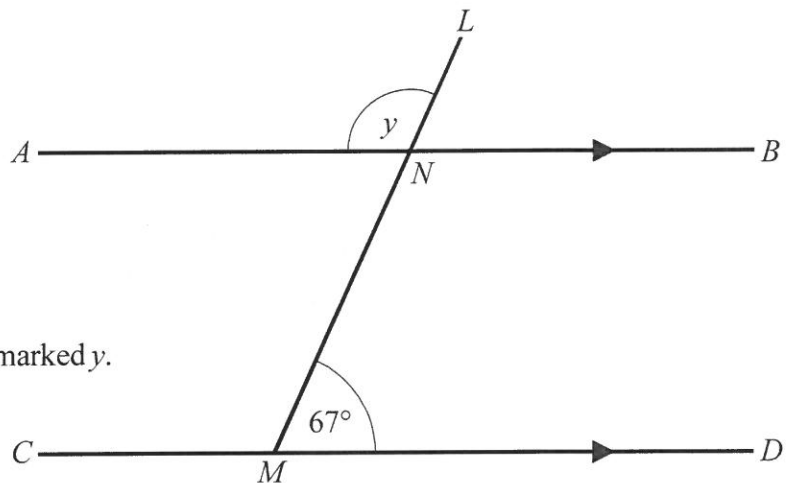
1)



(i) Write down the size of the angle marked a .

(ii) Give a reason for your answer.

2)



ANB is parallel to CMD .

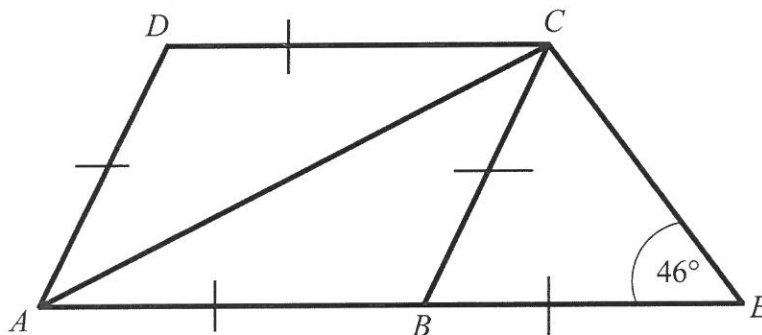
LMN is a straight line.

Angle $LMD = 67^\circ$

(i) Work out the size of the angle marked y .

(ii) Give reasons for your answer.

3)



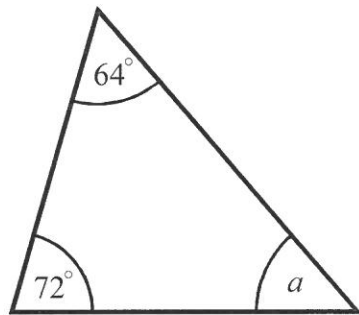
$ABCD$ is a rhombus.

BCE is an isosceles triangle.

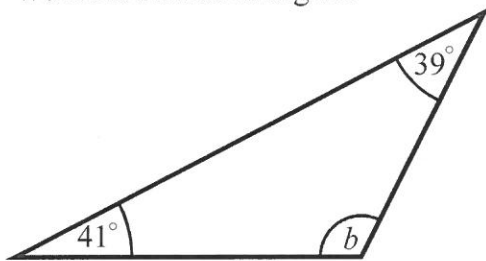
ABE is a straight line.

Work out the size of angle DCA .

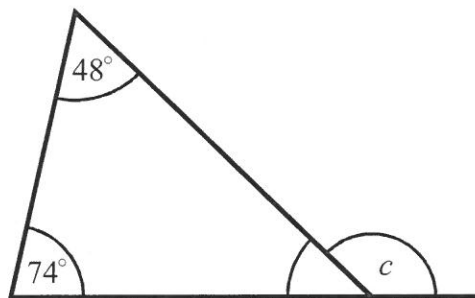
- 1) Work out the size of angle a .



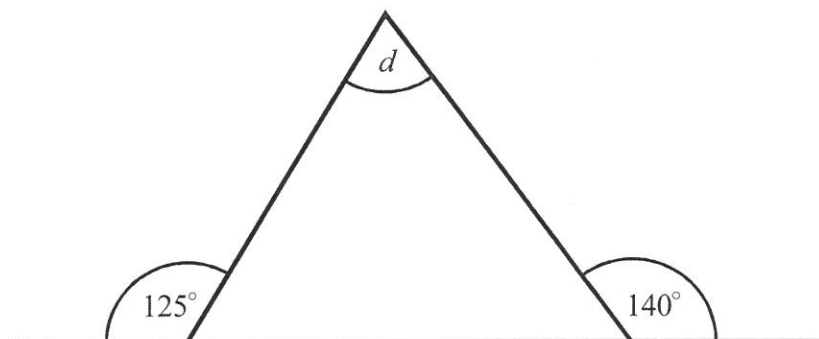
- 2) Work out the size of angle b .



- 3) Work out the size of angle c .



- 4) Work out the size of angle d .



Properties of Special Triangles

- 1) ABC is a triangle.

a) Find the size of angle A .

- b) Triangle ABC is equilateral.

Explain why.

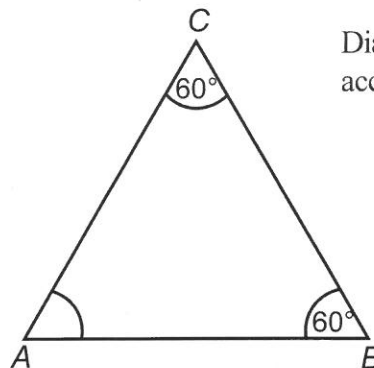


Diagram **NOT** accurately drawn

- 2) BCD is a triangle.

ABC is a straight line.

Angle $CBD = 70^\circ$.

$BD = CD$.

- a) (i) Work out the value of x .

(ii) Give a reason for your answer.

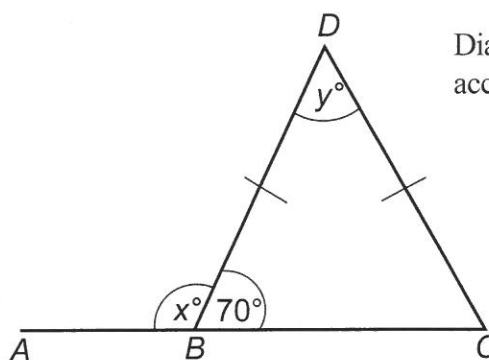


Diagram **NOT** accurately drawn

- b) (i) Work out the value of y .

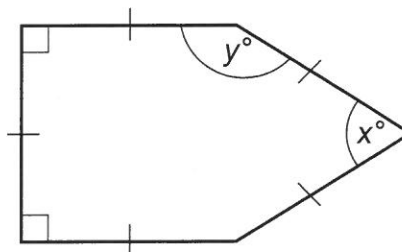
(ii) Give reasons for your answer.

- 3) The diagram shows a 5-sided shape.

All the sides of the shape are equal in length.

- a) (i) Find the value of x .

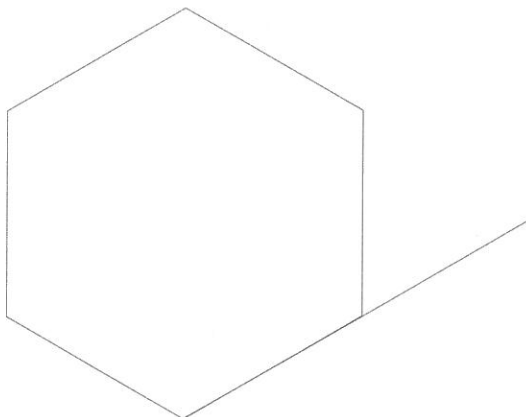
(ii) Give a reason for your answer.



- b) (i) Work out the value of y .

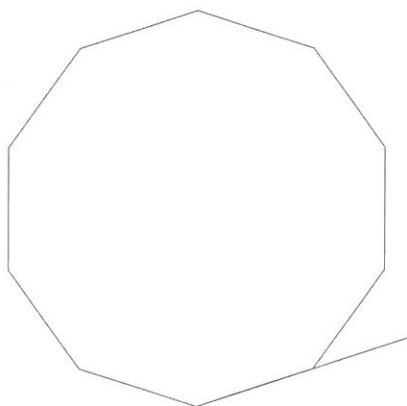
(ii) Explain your answer.

1)



- Work out the size of an **exterior** angle of a regular hexagon.
- Work out the size of an **interior** angle of a regular hexagon.

2)



- Name the regular polygon, above.
- Work out the size of an **exterior** angle and of an **interior** angle for this polygon.

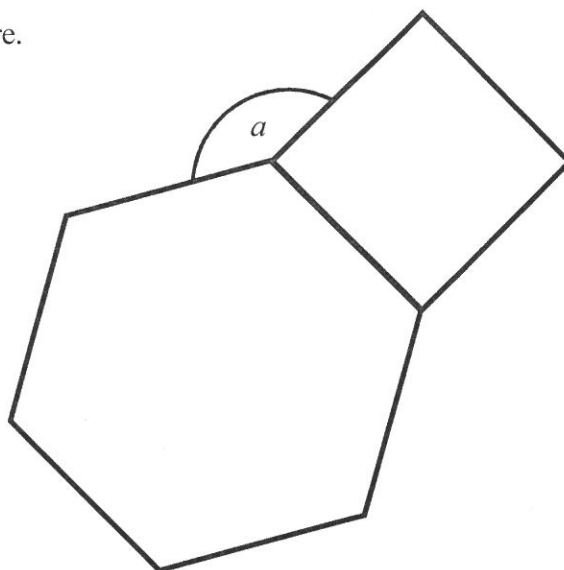
- The size of each **exterior** angle of a regular polygon is 40° .
Work out the number of sides of the regular polygon.

- The size of each **interior** angle of a regular polygon is 120° .
Work out the number of sides of the regular polygon.

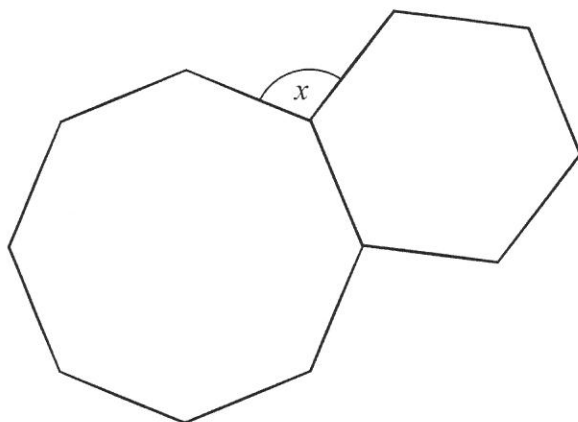


- 1) The diagram shows a regular hexagon and a square.

Calculate the size of the angle a .



- 2)



The diagram shows a regular octagon and a regular hexagon.

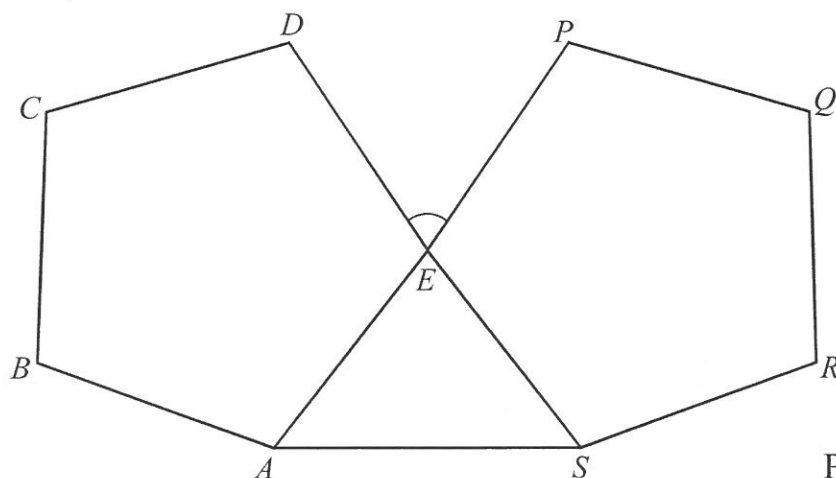
Work out the size of angle x .



- 3) $ABCDE$ and $PQRSE$ are regular pentagons.

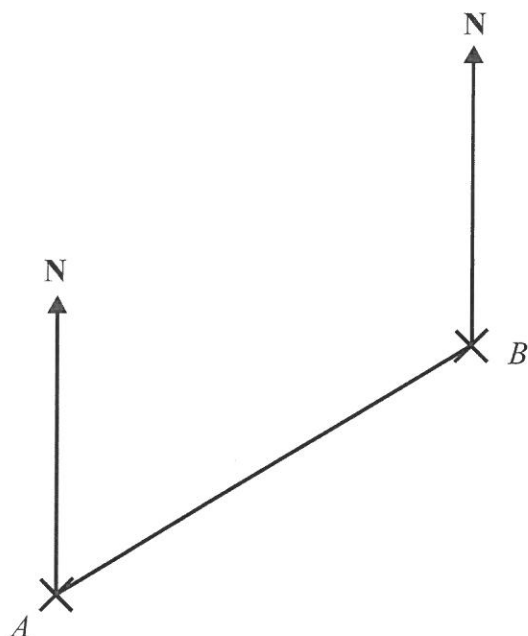
AES is an equilateral triangle.

Work out the size of angle DEP .



Bearings

- 1) The diagram shows the position of two telephone masts, A and B , on a map.

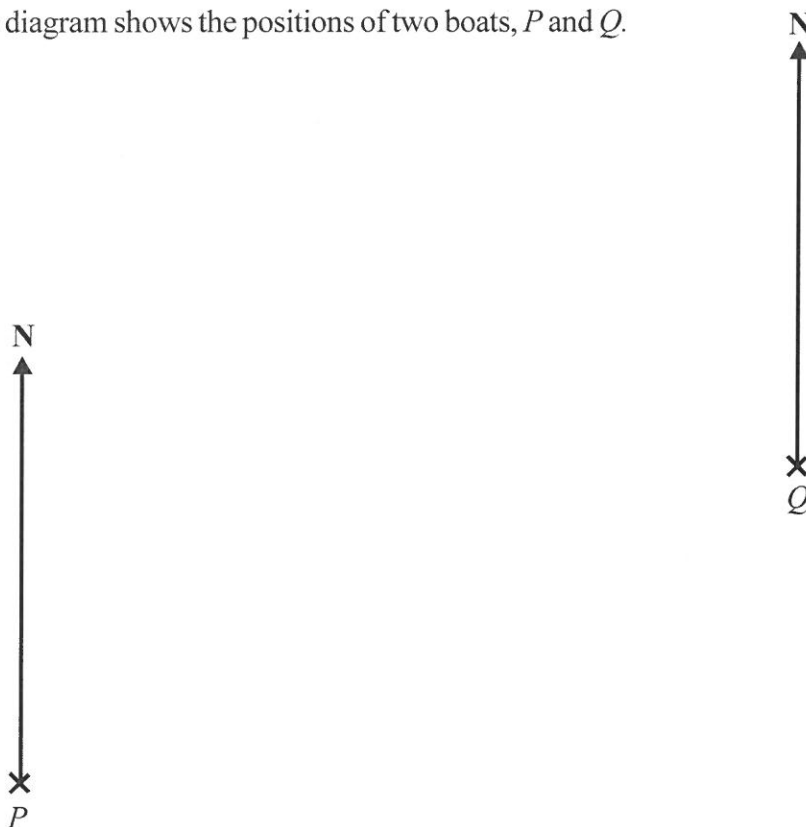


- a) Measure the bearing of B from A .

Another mast C is on a bearing of 160° from B .
On the map, C is 4 cm from B .

- b) Mark the position of C with a cross and label it C .

- 2) The diagram shows the positions of two boats, P and Q .



The bearing of a boat R from boat P is 050°

The bearing of boat R from boat Q is 320°

In the space above, draw an accurate diagram to show the position of boat R .
Mark the position of boat R with a cross ($\times$). Label it R .

Bearings

- 1) School B is due east of school A .
 C is another school.
 The bearing of C from A is 065° .
 The bearing of C from B is 313° .

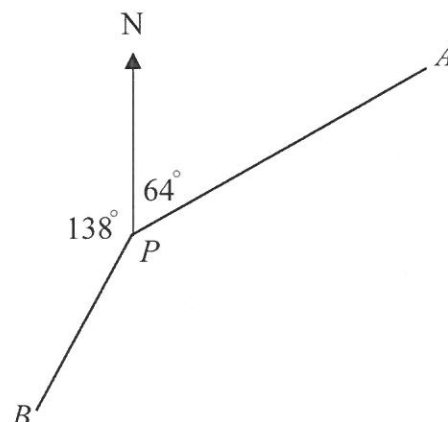
Complete the scale drawing below.
 Mark with a cross the position of C .



- 2) In the diagram, point A marks the position of Middlewich.
 The position of Middlemarch is to be marked on the diagram as point B .
 On the diagram, mark with a cross the position of B given that:
 B is on a bearing of 320° from A and
 B is 5 cm from A



- 3) **Work out** the bearing of
 a) B from P
 b) P from A



*Diagram NOT
accurately drawn.*

- 1) Ahmad does a statistical experiment.
He throws a dice 600 times.
He scores one, 200 times.
Is the dice fair? Explain your answer

- 2) Chris has a biased coin.
The probability that the biased coin will land on a tail is 0.3
Chris is going to flip the coin 150 times.
Work out an estimate for the number of times the coin will land on a tail.

- 3) On a biased dice, the probability of getting a six is $\frac{2}{3}$.
The dice is rolled 300 times.
Work out an estimate for the number of times the dice will land on a six.

- 4) On a biased dice, the probability of getting a three is 0.5
The dice is rolled 350 times.
Work out an estimate for the number of times the dice will land on a three.

- 5) Jenny throws a biased dice 100 times.
The table shows her results.

Score	Frequency
1	15
2	17
3	10
4	24
5	18
6	16

- a) She throws the dice once more.
Find an estimate for the probability that she will get a four.

- b) If the dice is rolled 250 times, how many times would you expect to get a five?

- 1) Two dice are rolled and their scores are multiplied together.
- a) Complete the possibility space to show all the possible results.

		First dice					
		1	2	3	4	5	6
Second dice	1						
	2						
	3				12		
	4						
	5						
	6			18			

- b) What is the probability of getting a result that is an even number?

- 2) Suppose there are three cards:

A **black card** that is black on both sides,

A **white card** that is white on both sides,

A **mixed card** that is black on one side and white on the other side.

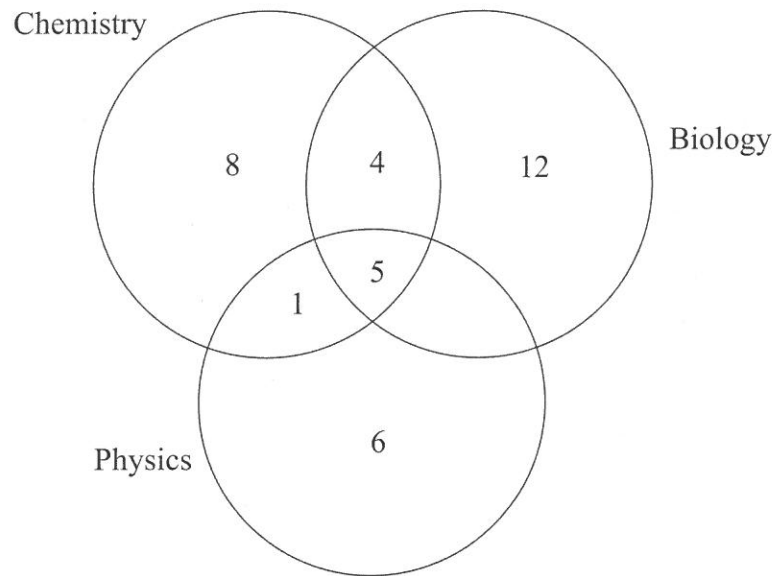
All the cards are placed into a hat and one is taken out at random.

It is placed on a table and the side facing up is black.

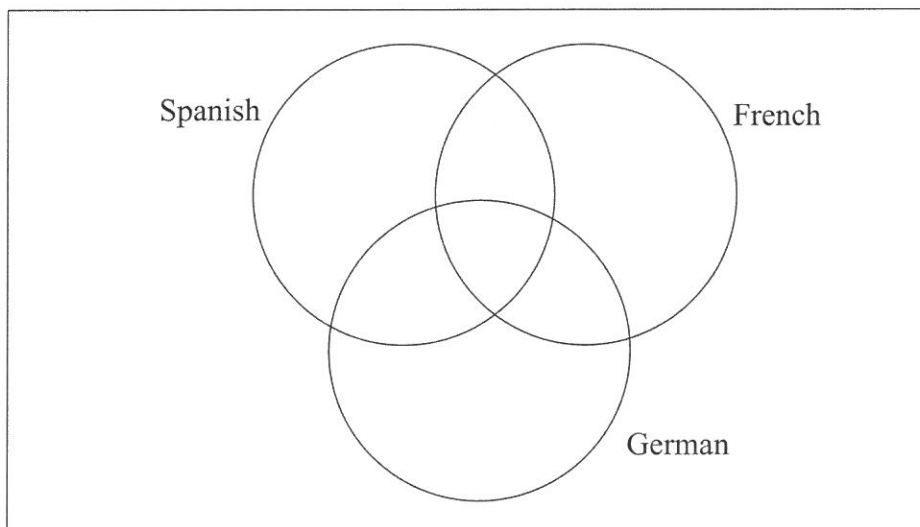
What is the probability that the other side of the card is also black?

Venn Diagrams

- 1) The Venn diagram shows the number of students studying one or more of the sciences Chemistry, Biology and Physics.



- a) How many students are represented in this Venn diagram?
- b) How many students are studying exactly two sciences?
- c) What is the probability that a student chosen at random is not studying Physics?
- 2) There are 31 student who study languages.
 All 31 study at least one of Spanish, French or German.
 4 study all three languages.
 9 study Spanish and French.
 7 study French and German.
 6 study Spanish and German.
 7 study only French and 5 study only German.
 Complete the Venn diagram to show this information.

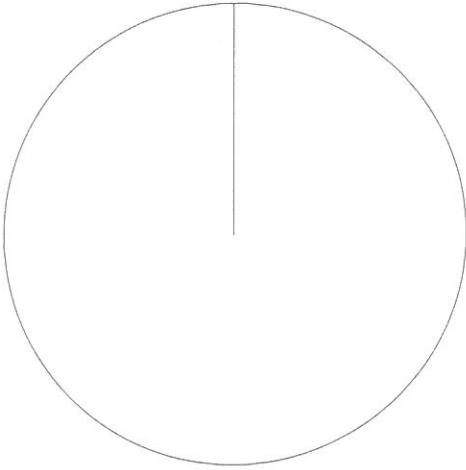


Pie Charts

- 1) Patrick asked some of his colleagues which was their favourite holiday destination.
The table shows the results.

City	Frequency
Alicante	8
Paris	7
Ibiza	15
St Lucia	1
Biarritz	9

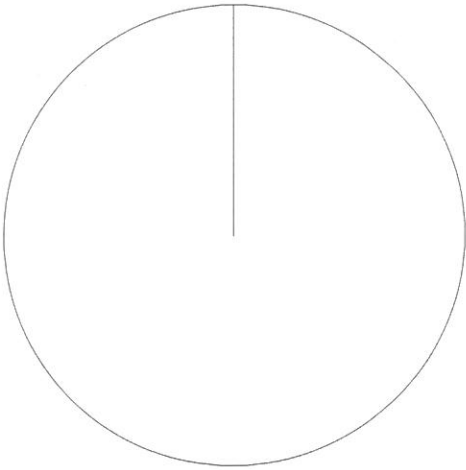
Draw a pie chart to illustrate the information.



- 2) Brian asked 60 people which region their favourite rugby team came from.
The table shows the results.

Region	Frequency
Southern England	9
London	23
Midlands	16
Northern England	12
Total	60

Draw a pie chart to illustrate the information.



- 3) Sophie represents her monthly expenses using a pie chart.

Diagram
accurately
drawn



Numbers from her table have been rubbed out by mistake.

Use the pie chart to complete the table.

		Angle
Clothes	£35	
Eating out		
Make up	£17	34°
Magazines		
Books		
Total	£180	

Scatter Diagrams

- 1) The scatter graph shows some information about the marks of six students. It shows each student's marks in Maths and Science.

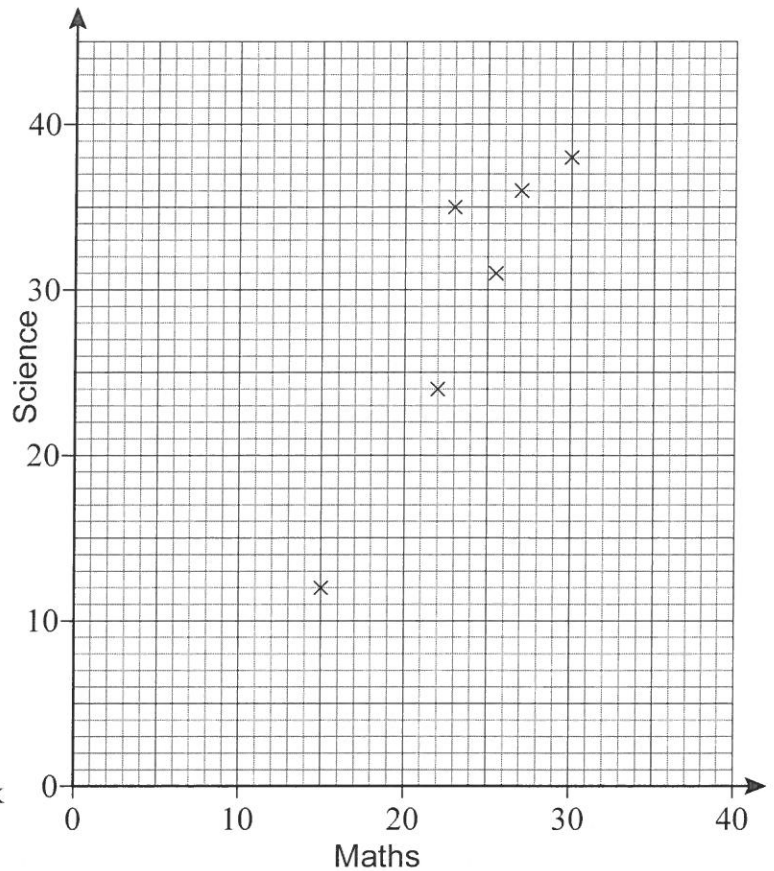
The table below shows the marks for four more students.

Maths	22	8	17	26
Science	30	12	24	24

- On the scatter graph, plot the information from the table.
- Draw a line of best fit.
- Describe the correlation between the marks in Maths and the marks in Science.

Another student has a mark of 18 in Science.

- Use the line of best fit to estimate the mark in Maths of this student.

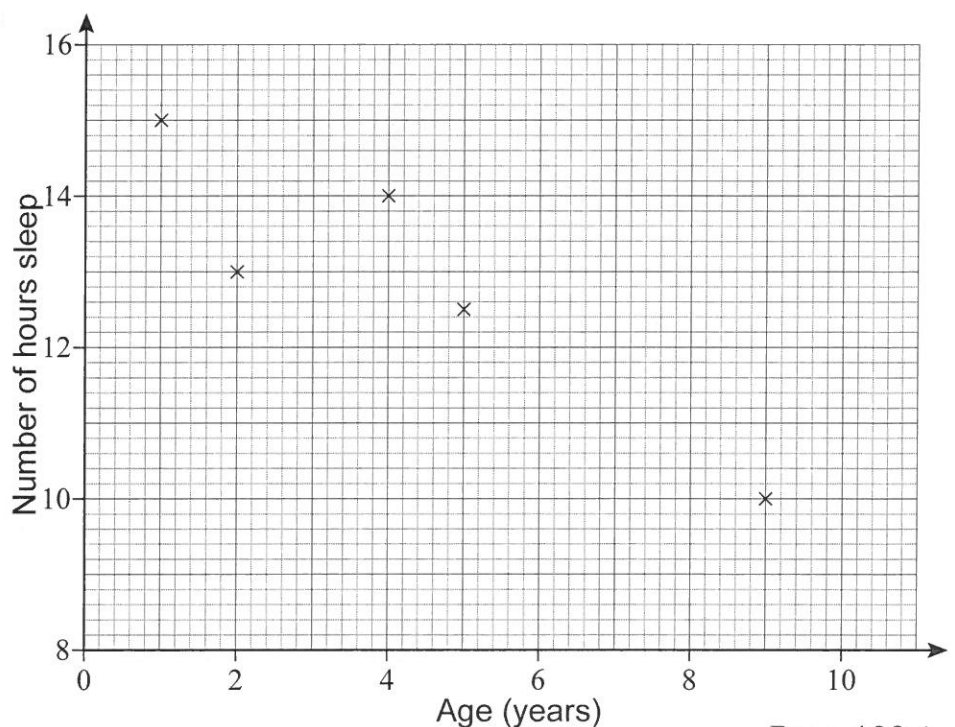


- 2) The table below shows the average daily number of hours sleep of 10 children.

Age (years)	4	2	5	1	9	6	8	7	10	1.5
Number of hours sleep	14	13	12.5	15	10	12.5	10.8	12	11	14

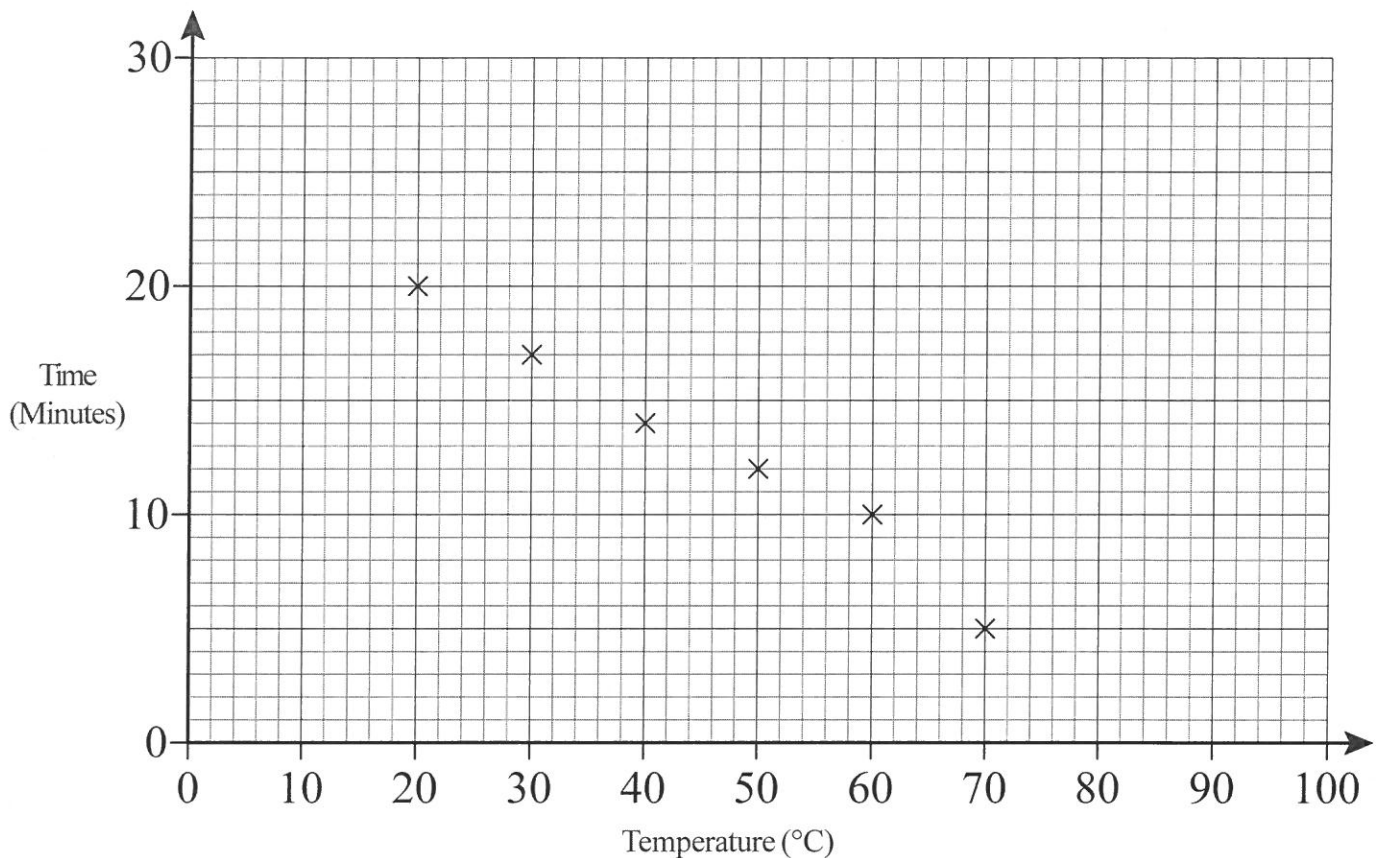
The first five results have been plotted on the scatter diagram.

- Plot the next five points.
- Draw a line of best fit.
- Describe the relationship between the age of the children and their number of hours sleep per day.
- Use your scatter graph to estimate the number of hours sleep for a 3 year old child.



Scatter Diagrams

- 1) Sue did an experiment to study the times, in minutes, it took 1 cm ice cubes to melt at different temperatures.
Some information about her results is given in the scatter graph.



The table shows the results from two more experiments.

Temperature (°C)	15	55
Time (Minutes)	21	15

- On the scatter graph, plot the results from the table.
- Describe the relationship between the temperature and the time it takes a 1 cm ice cube to melt.
- Find an estimate for the time it takes a 1 cm ice cube to melt when the temperature is 25 °C.

Sue's data cannot be used to predict how long it will take a 1 cm ice cube to melt when the temperature is 100 °C.

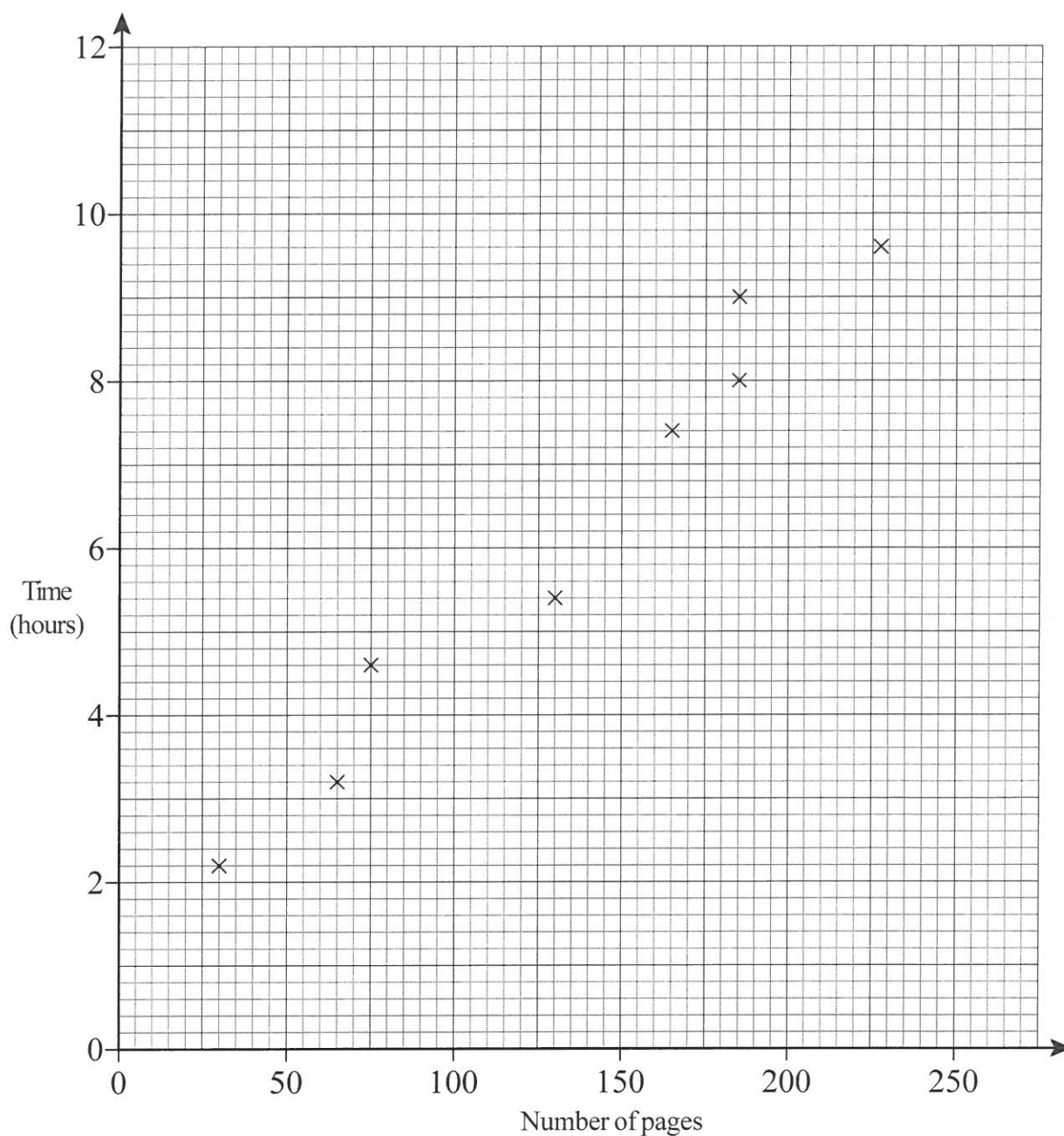
- Explain why.

Scatter Diagrams

- 1) Henry reads eight books.

For each book he recorded the number of pages and the time he took to read it.

The scatter graph shows information about his results.



- a) Describe the relationship between the number of pages in a book and the time Henry takes to read it.

Henry reads another book.

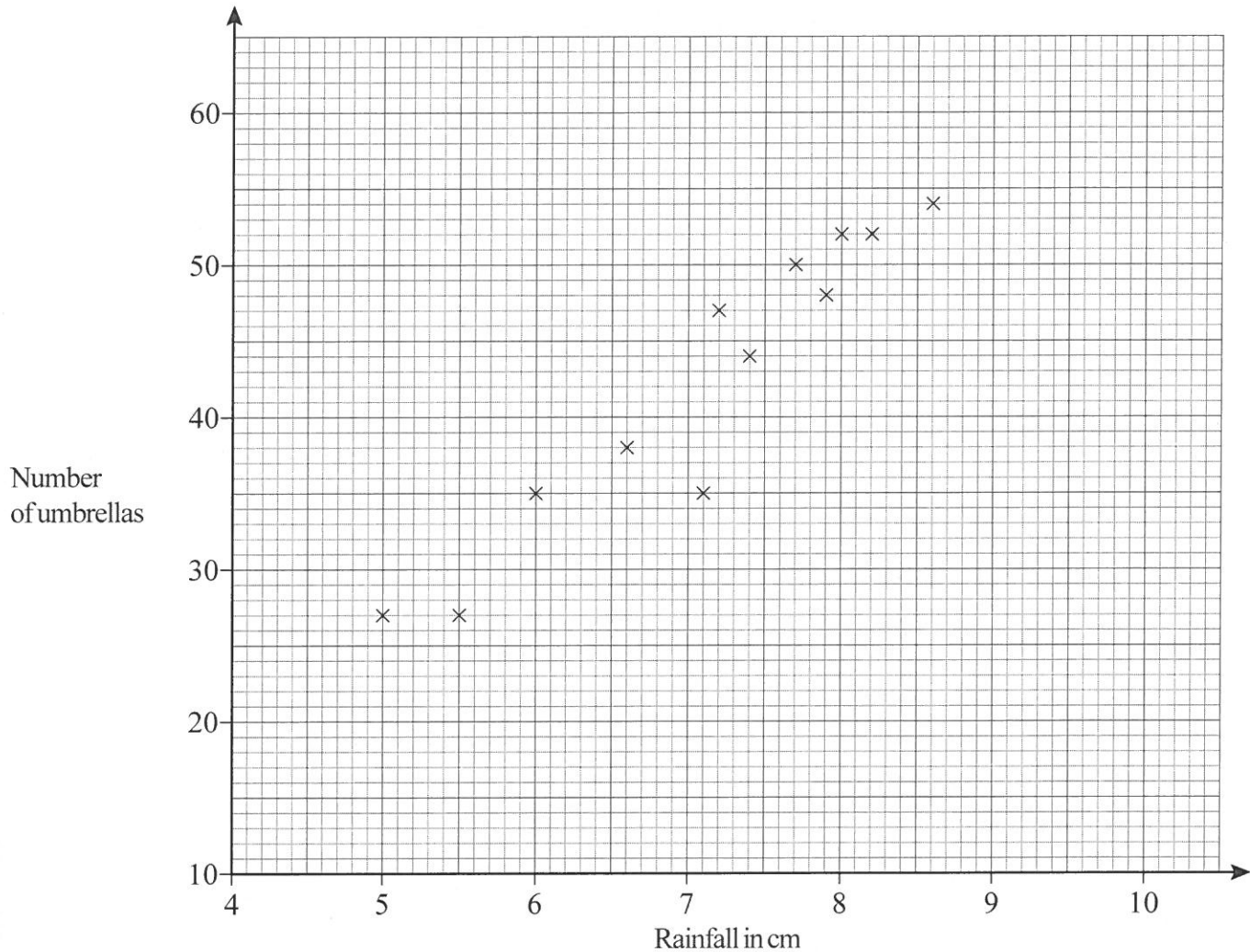
The book has 150 pages.

- b) Estimate the time it takes Henry to read it.

Scatter Diagrams

- 1) Mr Jones sells umbrellas.

The scatter graph shows some information about the number of umbrellas he sold and the rainfall, in cm, each month last year.



In January of this year, the rainfall was 6.2 cm.

During January, Mr Jones sold 32 umbrellas.

- Show this information on the scatter graph.
- What type of correlation does this scatter graph show?

In February of this year, Mr Jones sold 40 umbrellas.

- Estimate the rainfall for February.



- 1) The number of pens in each pupil's pencil case in a classroom has been counted. The results are displayed in a table.

Number of pens	Number of pupils
0	4
1	6
2	7
3	5
4	3
5	1

- Work out the total number of pens in the classroom.
- Write down the modal number of pens in a pencil case.
- Work out the mean number of pens in a pencil case.
- Work out the range of the number of pens in a pencil case.



- 2) Thomas is analysing the local football team. He records the number of goals scored in each football match in the past twelve months.

Thomas said that the mode is 7
Thomas is wrong.

- Explain why.
- Calculate the mean number of goals scored.

Goals scored	Frequency
0	7
1	5
2	3
3	6
4	2
5	1
6	1